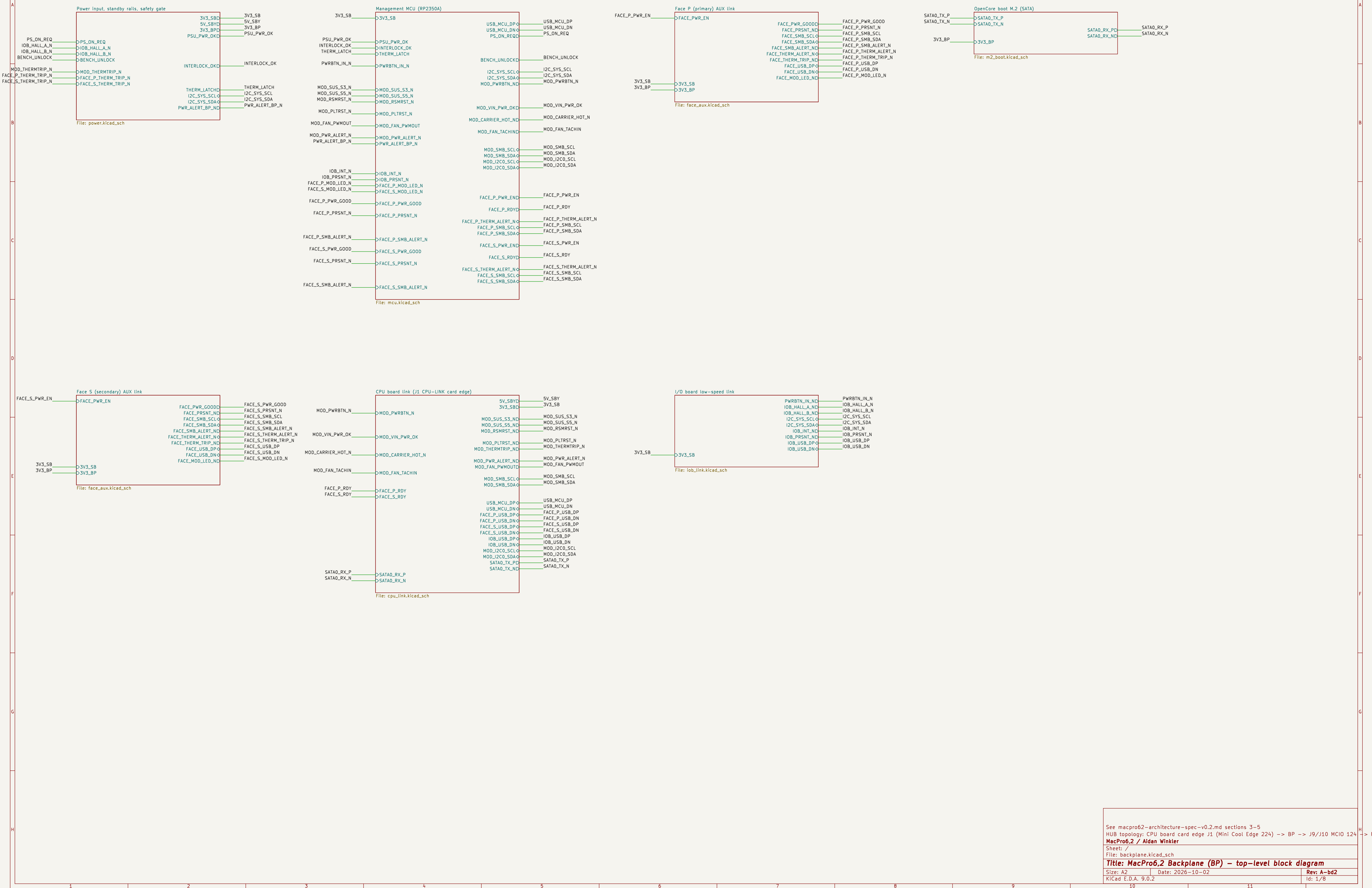


MP62 BACKPLANE rev A – BLOCK DIAGRAM. Sheets are stubs (labels only). Nets with the same label name are connected.

Off-board links: J1 CPU-LINK (card edge from CPU board) | J2 PSU-IN | J3/J4 Face P/S AUX (GH15) | J6 IOB-LINK (fan via IOB CONN_C, no J5) | J7 M.2 OpenCore | J8 SWD | J9/J10 MCIO 124 to Face P/S (PCIe hub)



3V3_SB_C	1	3V3_SB	
5V_SBY_C	2	5V_SBY	
3V3_BP_C	3	3V3_BP	
PSU_PWR_OK_C	4	PSU_PWR_OK	
PSU_ON_REQ_C	5	PSU_ON_REQ	
IOB_HALL_A_N	6	IOB_HALL_A_N	
IOB_HALL_B_N	7	IOB_HALL_B_N	
BENCH_UNLOCK_C	8	BENCH_UNLOCK	
INTERLOCK_OK_C	9	INTERLOCK_OK	
MOD_THERMTRIP_N	10	MOD_THERMTRIP_N	
FACE_P_THERM_TRIP_N	11	FACE_P_THERM_TRIP_N	
FACE_S_THERM_TRIP_N	12	FACE_S_THERM_TRIP_N	
THERM_LATCH_C	13	THERM_LATCH	
I2C_SYS_SCL_C	14	I2C_SYS_SCL	
I2C_SYS_SDA_C	15	I2C_SYS_SDA	
PWR_ALERT_BP_N	16	PWR_ALERT_BP_N	

BLOCK STUB (rev A block diagram). Contents planned:

- J2 PSU-IN Micro-Fit 3.0 2x4: 12V x2, GND x3, 11V_SB, PS_ON#, PWR_OK (stock PSU pinout TBD, adapter harness)
- 11V_SB -> 3V3_SB (MCU, EEPROMs, face aux in S5); 11V_SB -> 5V_SBY (COM-HPC suspend well)
- 12V_MAIN -> 3V3_BP buck (S0); input TVS + fuse; INA228 power monitor 0x40 on I2C_SYS, ALERT -> PWR_ALERT_BP_N (12V_MAIN / 11V_SB stay inside this sheet)
- BP 12 V load ~1 A (3V3_BP incl. Gen5 redrivers); fan moved to IOB CONN_C, CB / faces / IOB 12 V by their own PSU leads
- HW gate: PS_ON = INTERLOCK_OK AND PS_ON_REQ AND NOT THERM_LATCH (discrete logic, 3V3_SB)
- THERM_LATCH set by MOD_THERMTRIP_N / FACE_x_THERM_TRIP_N; cleared by AC cycle (TBD)
- INTERLOCK_OK = HALL_A AND HALL_B (or BENCH jumper AND MCU unlock)

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /Power input, standby rails, safety gate/
File: power.kicad_sch

Title: MP62 backplane: Power input, standby rails, safety gate

Size: A4	
KiCad E.D.A. 9.0.2	

Date: 2026-10-02

Id: 2/8





	1	2	3	4	5	6
A	5V_SBY◊ <u>4</u> 3V3_SB◊ <u>2</u> MOD_PWRBTN_ND◊ <u>3</u> MOD_SUS_S3_N◊ <u>4</u> MOD_SUS_S5_N◊ <u>5</u> MOD_RSMRST_N◊ <u>6</u> MOD_VIN_PWR_OK◊ <u>7</u> MOD_PLTRST_N◊ <u>8</u> MOD_THERMTRIP_N◊ <u>9</u> MOD_CARRIER_HOT_N◊ <u>10</u> MOD_PWR_ALERT_N◊ <u>11</u> MOD_FAN_PWMOUT◊ <u>12</u> MOD_FAN_TACHIN◊ <u>13</u> MOD_SMB_SCL◊ <u>14</u> MOD_SMB_SDA◊ <u>15</u> FACE_P_RDY◊ <u>16</u> FACE_S_RDY◊ <u>17</u> USB_MCU_DP◊ <u>18</u> USB_MCU_DN◊ <u>19</u> FACE_P_USB_DP◊ <u>20</u> FACE_P_USB_DN◊ <u>21</u> FACE_S_USB_DP◊ <u>22</u> FACE_S_USB_DN◊ <u>23</u> IOB_USB_DP◊ <u>24</u> IOB_USB_DN◊ <u>25</u> MOD_I2C0_SCL◊ <u>26</u> MOD_I2C0_SDA◊ <u>27</u> SATA0_TX_P◊ <u>28</u> SATA0_TX_N◊ <u>29</u> SATA0_RX_P◊ <u>30</u> SATA0_RX_N◊ <u>31</u>	5V_SBY 3V3_SB MOD_PWRBTN_N MOD_SUS_S3_N MOD_SUS_S5_N MOD_RSMRST_N MOD_VIN_PWR_OK MOD_PLTRST_N MOD_THERMTRIP_N MOD_CARRIER_HOT_N MOD_PWR_ALERT_N MOD_FAN_PWMOUT MOD_FAN_TACHIN MOD_SMB_SCL MOD_SMB_SDA FACE_P_RDY FACE_S_RDY USB_MCU_DP USB_MCU_DN FACE_P_USB_DP FACE_P_USB_DN FACE_S_USB_DP FACE_S_USB_DN IOB_USB_DP IOB_USB_DN MOD_I2C0_SCL MOD_I2C0_SDA SATA0_TX_P SATA0_TX_N SATA0_RX_P SATA0_RX_N	BLOCK STUB (rev A block diagram). Contents planned: – J1: Amphenol Mini Cool Edge 224 (ME1022410103011) vertical SMT; MP62 hub pinout (docs/cpulink_224_pinout_draft.csv, CB mapping column) – PCIe HUB: J1 x16 -> J9 MCIO 124 (Face P), J1 x4 -> J10 MCIO 124 (Face S), REFCLK P/S; 85 ohm L1 (TX) / L6 (RX) – Optional Gen5 build: U10–U13 (Face P) + U14 (Face S) TI DS320PR810 linear redrivers + 220 nF output AC caps – Gen4 build: no redrivers, no AC caps on BP (host–TX caps on module/CPU board, device–TX caps on face) – PWR_ALERT# (A–row bay 4, was RSVD_LS1): CB INA228 0x45 ALERT + U11 FLT#, OD, BP 10k pull–up (live power target) – Also carries: power/state sideband (COM–HPC names, CB = Z790 PCH + EC), fan PWMOUT/TACHIN (demand only), SMBus, I2C0, USB2 x4, SATA0, 5V_SBY x6, 3V3_SB x2, GND x80 – USB2 x4 = MCU (BP MCU), FACEP / FACES (face AUX pins 13/14), SPARE -> IOB–LINK J6 13/14 (IOB hub H2: USB–A A1–A4 + Bluetooth) – BP makes PERST#_x = PLTRST# AND FACE_x_RDY (MCU asserts after FACE_x_PWR_GOOD); CLKREQ#_x end on BP; WAKE#_x ORed to WAKE0# – 12 V main for the CPU board does NOT pass through J1 (CB LUG1/LUG2 single 12 V entry, U11 eFuse)			
B						
C						
D						

Block stub: hierarchical labels + one non–BOM stub symbol (#BLKn, passive pins) so ERC sees the n			
Sheet: /CPU board link (J1 CPU–LINK card edge)/ File: cpu_link.kicad_sch			
Title: MP62 backplane: CPU board link (J1 CPU–LINK card edge)			
Size: A4	Date: 2026–10–02	Rev: A–bd2	
KiCad E.D.A. 9.0.2		Id: 6/8	

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC can be used.

Sheet: /I/O board low-speed link/
File: iob_link.kicad_sch

Title: MP62 backplane: I/O board low-speed link

Size: A4	Date: 2026-10-02	Rev: A-bd2
KiCad E.D.A. 9.0.2	Id: 7/8	

- J6 JST GH 15P (BM15B vertical) <-> IOB J6, 1:1 GH 15P cable (ICD 2026-10-02, pinout = IOB plan 5.3):
- 1,2 3V3_SB (BP -> IOB) | 3,7,12,15 GND | 4 PWRBTN_IN_N | 5 HALL_A_N | 6 HALL_B_N | 8 I2C_SYS_SCL | 9 I2C_SYS_SDA
- 10 IOB_INT_N (OD, IOB -> BP, BP pull-up; PD IRQs + LIS2DH12 + IOB INA228 0x41 ALERT) | 11 IOB_PRSNT_N (IOB ties to GND; BP 10k pull-up to 3V3_SB) | 13/14 USB2_LINK D+/D- (CPU-LINK USB2_SPARE)
- IOB high-speed (USB3 x10, DDI, 2x i226 PCIe, VBAT) runs CB J3 -> IOB HS1 directly; 12 V via the IOB 12P PSU header, not the BP

